Jie “JW” Wu, Ph.D.

Assistant Professor, Department of Computer Science
Michigan Technological University, Houghton, MI
jie.jw.wu@mtu.edu | https://jie-jw-wu.github.io

Education

Ph.D. in Systems Engineering, George Washington University (advisors: Thomas Mazzuchi, Shahram Sarkani) — May 2023

M.S. in Computer Science, Shanghai Jiao Tong University (advisor: Liqing Zhang) — Mar 2015

B.S. in Computer Science (ACM Honored Class), Shanghai Jiao Tong University — May 2012

Appointments

Assistant Professor, Department of Computer Science, Michigan Technological University, Houghton, MI — Aug 2025 – present

Postdoctoral Fellow, University of British Columbia (with Prof. Fatemeh Fard), B.C., Canada — Dec 2023 – Aug 2025

Software Engineer, Snap Inc. (Snapchat), Seattle, WA: Friend Recommendation, Core Growth ML, Friend Infrastructure — Feb 2017 – Jul 2023

Software Engineer, Microsoft Corporation (OneNote Canvas & Ink), Redmond, WA — Oct 2015 – Feb 2017

Earlier: Software Engineer, ArcSite (2015); Research Intern, Google Research (2014) and Microsoft Research Asia (2011–2014) — 2011 – 2015

Research Interests

Intent-aware software engineering: AI-assisted techniques that align AI behavior with developer intent and help humans understand, guide, and verify AI-generated software. Keywords: AI for software engineering, software engineering for AI, agentic AI, large language models for code.

Refereed Publications

Journal articles

[J1] **Jie JW Wu***, Ayanda Patrick Herlihy*, Ahmad Saleem Mirza, Ali Afoud, Fatemeh Fard. GrowthHacker: Automated Off-Policy Evaluation Optimization Using Code-Modifying LLM Agents. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026. (*equal contribution)

[J2] Meghdad Dehghan, **Jie JW Wu**, Fatemeh H. Fard, Ali Ouni. MergeRepair: An Exploratory Study on Merging Task-Specific Adapters in Code LLMs for Automated Program Repair. *Empirical Software Engineering*, 2026.

[J3] Sathvik Joel*, **Jie JW Wu***, Fatemeh H. Fard. A Survey on LLM-based Code Generation for Low-Resource and Domain-Specific Programming Languages. *ACM TOSEM*, 2025. (*equal contribution; journal-first presentation at FSE 2026)

[J4] **Jie JW Wu**, Fatemeh H. Fard. Benchmarking the Communication Competence of Code Generation for LLMs and LLM Agent (HumanEvalComm). *ACM TOSEM*, 2025. (journal-first presentation at FSE 2026)

[J5] **Jie JW Wu**, Thomas A. Mazzuchi, Shahram Sarkani. A Multi-Objective Evolutionary Approach Towards Automated Online Controlled Experiments. *Journal of Systems & Software*, 2023.

[J6] **Jie JW Wu**, Thomas A. Mazzuchi, Shahram Sarkani. Comparison of Multi-Criteria Decision-Making Methods for Online Controlled Experiments in a Launch Decision-Making Framework. *Information and Software Technology*, 2023.

Conference papers

[C1] Mohamad Khajezade, **Jie JW Wu**, Fatemeh H. Fard, Gema Rodríguez-Pérez, Mohamed Sami Shehata. Investigating the Efficacy of Large Language Models for Code Clone Detection. *ICPC 2024*, Early Research Achievements.

[C2] **Jie JW Wu**. AutoOffAB: Toward Automated Offline A/B Testing for Data-Driven Requirement Engineering. *FSE 2024*, Ideas, Visions and Reflections.

[C3] **Jie JW Wu**. An Exploratory Study of V-Model in Building ML-Enabled Software: A Systems Engineering Perspective. *CAIN 2024*. **Distinguished Paper Award Candidate.**

[C4] Amal Akli, **Jie JW Wu**, Cedric Richter, Mike Papadakis. The Clarification Challenge: Towards Coding Assistants that Clarify Unclear User Requirements. *ICSE 2027*, Competition Track.

[C5] Jie Wu, Changhu Wang, Liqing Zhang, Yong Rui. Offline Sketch Parsing via Shapeness Estimation. *IJCAI 2015*.

[C6] Jie Wu, Changhu Wang, Liqing Zhang, Yong Rui. Sketch Recognition with Natural Correction and Editing. *AAAI 2014*. (Demonstrated at Microsoft TechFest 2015)

[C7] Jie Wu, Liqing Zhang. Gestalt Saliency: Salient Region Detection Based on Gestalt Principles. *IEEE ICIP 2013*.

Workshop papers (peer-reviewed)

[W1] Ali Afoud (advisee), **Jie JW Wu**. Verification as an Architectural Layer for LLM Agents: A V-Model Design, and a Pilot Study of Its Deterministic Core. *JAW 2 Workshop at ASE 2026*.

[W2] Rishab Sharma, Iman Saberi, Elham Alipour, **Jie JW Wu**, Fatemeh H. Fard. FISCAL: Financial Synthetic Claim–document Augmented Learning for Efficient Fact-Checking. *NeurIPS 2025 Workshop on GenAI in Finance*.

[W3] **Jie JW Wu**. Large Language Models Should Ask Clarifying Questions to Increase Confidence in Generated Code. *MAPS 2023 (at FSE)*.

Non-archival (reviewed and accepted; not in proceedings)

[N1] **Jie JW Wu**, Feiyu E, Bo Chen. Metamorphic Testing for Clinical ML Models: A Framework Proposal and Pilot Study. *AIware 2026, arXiv Track* (presented in Montreal, Canada), 2026.

Preprints under review

[P1] **Jie JW Wu**. AssumptionMiner: Extracting, Tracing, and Revising Implicit Assumptions in LLM Code Generation. arXiv:2607.22898, 2026.

[P2] **Jie JW Wu**, Manav Chaudhary, Davit Abrahamyan, Arhaan Khaku, Anjiang Wei, Fatemeh H. Fard. Can Code Language Models Learn Clarification-Seeking Behaviors? (ClarifyCoder) arXiv:2504.16331, 2025.

[P3] Qiang Han, **Jie JW Wu**, Bo Chen. Overthink-Triggered Slowdown Attacks on LVLM-Based Robotic Systems. arXiv:2607.01518, 2026.

[P4] Shawn Zhao, Fatemeh H. Fard, **Jie JW Wu**. MANTRA: A Framework for Multi-stage Adaptive Noise Treatment During Training. arXiv:2512.04319, 2025.

Patents

- US 11,171,972 B1, Additive Friend Suggestion for Account Security (with J. Brody, D. Giovannini, E. Koai, L. Zhong; Snap Inc.).
- US 9,940,513 B2, Intuitive Selection of a Digital Stroke Grouping (with C. R. Iskander, I. W. Mikutel, S. E. Sykes, D. G. Garber; Microsoft).

Teaching

- **Michigan Tech (instructor):** CS 5090 Software Engineering with Generative AI, graduate (Fall 2026); CS 4713 Embedded Software Foundations, two sections, flipped classroom with in-class active learning (Fall 2026); CS 3712 Software Quality Assurance, 62 students (Spring 2026); CS 4321 Introduction to Algorithms, 82 students (Fall 2025).
- **University of British Columbia (instructor):** COSC 320 Analysis of Algorithms, 80 students (Fall 2024). Instructional Skills Workshop (ISW) certification, 24-hour program, 2024.
- **Shanghai Jiao Tong University:** Head Teacher, ACM Honored Class of 2016 (2012–2013); Teaching Assistant, Practice on Programming and Coding Ability (Fall 2011).

Advising and Mentoring

- **PhD students, Michigan Tech:** Joshana Shakya (advisor, Fall 2026–); Indrajeet Roy (advisor, Fall 2026–); Josh Dafoe (co-advisor with Prof. Bo Chen, Summer 2026–); Qiang Han (research co-mentor, Prof. Bo Chen's group, Fall 2025; [P3]).
- **Undergraduate researchers, Michigan Tech:** Andrew Sadler (CC-REU Fellow, Summer 2026–); Gabe Dautovi (URS Scholar, Summer 2026–).
- **UBC and other (2023–2025):** Ali Afoud (B.S., advisor; first author of [W1]); Ayanda Patrick Herlihy (Birmingham City University, mentor; co-first author of [J1]); Shawn Zhao (PhD, mentor; [P4]); Meghdad Dehghan (M.Sc., mentor; [J2]); Dabira Omotoso (honors thesis co-supervisor); Ahmad Saleem Mirza (mentor; [J1]); Manav Chaudhary (IIIT Hyderabad, Mitacs intern, mentor; [P2]); Abhinav Thota (published honors thesis, mentor); Md Jumar Alam (published M.Sc. thesis, mentor).
- **Committees:** PhD proposal defense committee, Reiki Tanaya Karuka (advisor: Prof. Vinh Nguyen), 2026; PhD qualifying exam committees, Qiang Han and Zhuoyan Xu (advisor: Prof. Bo Chen), 2026; M.S. thesis committee, Connor Fleser (advisor: Prof. Charles Wallace), 2027; PhD advisory committee, Ajay MNV Penmatsa (advisor: Prof. Neerav Kaushal).

Selected Professional Service

- **Organizing:** Benchmark and Dataset Co-chair, ICSME 2027; Publicity and Social-Media Co-chair, CAIN 2027; Facilitator and Mentor, Digital Research Infrastructure (DRI) Sensitization Workshop, UBC Okanagan.
- **Program committees:** ICSE 2027 (Research Track); CAIN 2027; AIware 2025 (Benchmark and Dataset Track); RAIE 2025; CAIN 2025; SANER 2025 (Tool and Demo Track); ASE 2024 (Artifact Evaluation Track); FSE 2024 (Posters Track).
- **Journal reviewer:** ACM TOSEM (2025, 2026); Communications of the ACM (2026); IEEE Software (2026); IEEE Transactions on Software Engineering (2025); IEEE Transactions on Dependable and Secure Computing (2025); Information and Software Technology (2025); Journal of Systems and Software (2024, 2025).

• **Michigan Tech:** Software Engineering Curricula Committee, Department of Computer Science.

Selected Talks

- FSE 2026, Montreal (journal-first): “A Survey on LLM-based Code Generation for Low-Resource and Domain-Specific Programming Languages” and “Benchmarking the Communication Competence of Code Generation for LLMs and LLM Agent”; AIware 2026: “Metamorphic Testing for Clinical ML Models.”
- Michigan Tech CS Colloquium (2025): “Clarification-Aware Code Generation: Benchmarks, Techniques, and Open Problems”; PNW PLSE 2025, Uber Engineering, Seattle: “Clarification-Aware Code Generation”; Washington State University and Michigan Tech (2025): “Building Trustworthy and Safe AIware With Systems Thinking”; MAPS 2023 at FSE, San Francisco: “Large Language Models Should Ask Clarifying Questions.”
- Training session, DRI sensitization workshop, UBC Okanagan (2024): “Optimizing Large Language Model Workflows on ARC”; guest lecture, CS 1000, Michigan Tech (2025); learning sessions at Snap Inc. (2021, 2022).

Awards and Honors

Travel Grant Award, FSE 2024 ($1k)	Jul 2024
Distinguished Paper Award Candidate, CAIN 2024	Apr 2024
Travel Grant Award, ICSE 2024 ($1.2k); Travel Grant Award, FSE 2023 ($1.7k)	2023–2024
Engineer of the Month (two-time winner), Friending Team, Snap Inc., for contributions to candidate generation, data, and privacy of the Friending system	Jan 2022
3rd place, Infrastructure Cost Camp, Snap Inc. ($800k/year cost saving)	Jun 2020
Letter from the CEO, Snap Inc., in recognition of the new friend-suggestion model	Jun 2019
National Scholarship (highest level), China	Oct 2014
Excellent Master Academic Scholarship (1st class), SJTU; Intel Fellowship	2013
Top 1% Excellent Undergraduate Thesis, SJTU (38 of 4,000 theses)	Jun 2012
Microsoft Fellowship for Undergraduates (30 recipients in China)	Aug 2011
First Class Prize, National Olympiad in Informatics in Provinces (NOIP)	Oct 2007